Research Methodology & IPR

Debesh Choudhury

Lecture Number 03 (November 17, 2025, at 2:30 PM)

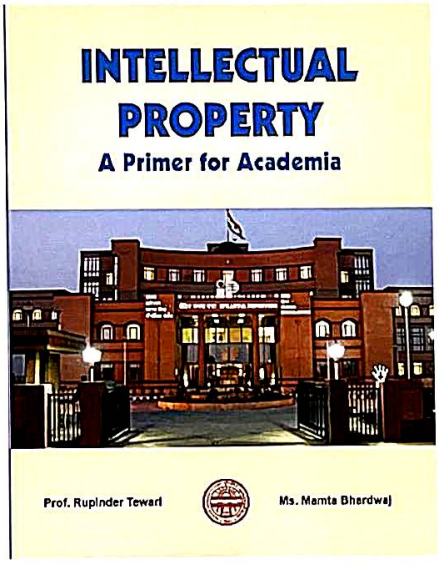
Intellectual Property $\xleftrightarrow{\text{Claiming ownership}}$ Patents, Trademarks, Designs, Copyright

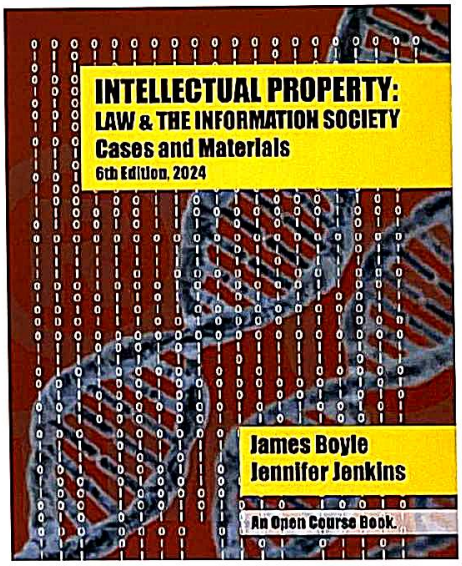
A Course for M.Tech. in Biomedical Instrumentation, Computer Science & Engineering, and Optics & Optoelectronics

University of Calcutta, July - December 2025 Semester

1 Overview
2 IP
3 Patent
4 Copyright
5 Trademark
6 Design
7 TK
8 Patenting
9 WIPO
10 Questions?

Reference books





Aspects of Intellectual Property

- Intellectual Property
- Patents
- Copyrights
- Designs
- Trademarks

What is intellectual property or IP?

- Intellectual Property (IP) is a special category of property created by human intellect (mind) in the fields of arts, literature, science, engineering, trade, etc.
- IP is intangible (i.e. invisible and indivisible) in nature and differs from the tangible property, such as land, house, car, gold, etc.
- Intellectual Property Rights (IPR) are the privileges accorded to the creator/inventor (of IP) in conformance with the laws.
- These rights are given to the creator/inventor in exchange for revealing the process of creation/invention in the public domain.
- The inventor is conferred with the special rights to use, sell, distribute, offering for sale and restricting others from using the invention without their prior permission.
- IP comprises of two branches i.e. Copyrights and Related Rights and Industrial Property Rights.
- Copyrights refer to the creative expressions in the fields of literature and art, such as books, publications, architecture, music, etc.
- The Industrial Property Rights refer to the Patents, Trademarks, Trade Services, Industrial Designs and Geographical Indications.

Impact of IP in the Economic and Cultural Development

- The economic and social development of a society is dependent on creativity, and hence on IPR.
- The protection provided by the IPR to the creators/innovators is an act of incentivization for encouraging them to create more and motivates others to create new and novel things.
- But, compliance with the Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement has affected the farming community as they are unable to store seeds for the next crop.
- To circumvent the negative impact of IPR, certain laws, exceptions and limitations associated with IPR have been enacted to maintain a balance and empower people with special rights.
- The use of copyrighted material for education and religious ceremonies is exempted from the operation of the rights granted in the Copyright Act.
- A patent can be revoked in favour of compulsory licensing by the government during an emergency or a natural calamity.
- Cloning of human embryos is banned for IP protection, and so is the creation of super microbial pathogens, which can play havoc with human lives.
- The government should protect Traditional Knowledge (TK) and genetic resources, which are rapidly coming under the governance of sometimes conflicting IPR policies.

Patents

- A patent is an exclusive right granted for an innovation that generally provides a new way of doing something or offers a new technical solution to a problem.
- The first patent in India was granted to George Alfred DePenning from Calcutta, for his invention, 'An Efficient Punkah Pulling Machine' in 1856.
- The "Priority Date" is the date on which the first application for the invention is filed, whether it is provisional or with complete specifications.
- Amendments of Indian patents act permitted enhancing the term of the patent from 14 years to 16, and further to 20 years.
- It also permits filing of Provisional Application and submission of Complete Application within 9 (extended up to 12) months from the date of filing the application.
- Food, medicine and surgical and curative devices should be made available to the masses at the cheapest rate by giving reasonable compensation to the owner of the patent.
- India is a member of all prominent Conventions and Treaties related to the facilitation of the inventors for international filing and protecting the rights over the inventions globally.

Copyright and related rights

- The concept of copyrights started way back in the 15th century. However, the actual need for copyrights law was felt only after the invention of printers and copiers.
- The evolution of copyrights law in India occurred in three phases: First two phases were by the British (1847 and 1914).
- The Indian Copyright Act 1957 was enacted with six amendments thereafter (1983, 1984, 1992, 1994 and 1999, 2012).
- Most of the amendments in copyright laws were in the digital environment.
- India is an active member of nearly all significant international Conventions/Treaties related to Copyright Law.
- For book publishing, there is ISBN (International Standard Book Number), for periodicals, there is ISSN (International Standard Serial Number).
- For music publishing, there is ISMN (International Standard Music Number). Unfortunately, India has no representation in the global ISMN organization. (I have personally checked it).

Trademarks

- A trademark is a word, phrase, symbol, design, or combination of these elements used to identify and distinguish the goods or services of one business from those of others, indicating their source.
- The first statutory law related to Trademarks (TM) in India was the Trade Marks Act, 1940, which was carved out from the Trade Marks Act, 1938 of the UK.
- It was followed by the incorporation of provisions of TM stated in the Indian Penal Code, Criminal Procedure Code and the Sea Customs Act.
- Later on, Trade Marks Act, 1940 was rechristened as Trade and Merchandise Marks Act, 1958.
- Nearly four decades later, this Act was repealed by the Trade Marks Act, 1999. It is the current governing law related to registering TM.

Various Design IPs

- A few years after the Indian Patent Act was enacted, it was felt that Designs could also pass the criteria of the invention and thus should be included in the Patent Act.
- In the 21st century, Information Technology (IT) has revolutionized, and a new class of IP called Layout-Design of the Semiconductor Integrated Circuits evolved.
- International organizations, including WTO and TRIPS Agreement laid down rules and regulations regarding the protection of Semiconductor Integrated Circuits and Layout Designs (SICLD).
- India being a member of the WTO also passed an Act called the SICLD Act, 2000.
- Another IP called Industrial Designs was recognized in the 18th century and the Indian legislation enacted the Patterns and Designs Act in 1872 for the first time.
- The Act was enacted to protect the rights over the creation of the designs and novel patterns by the inventors.
- In the year 2000, a dedicated Act for the ID was passed, which came into force in 2001.

Traditional Knowledge (TK)

- It is the ancient and indigenous knowledge held by any community or a group of people.
- In olden times it was not recorded anywhere and was available only in oral form. So, Traditional Knowledge (TK) was verbally passed on to future generations.
- Though there is no official record but some forms of TK find appearance in the culture, stories, legends, folklore, rituals, songs, etc.
- Previously, there was no mechanism available to protect TK, but now, it has been recognized as IPR under an agreement.
- The Government of India has created a digital library termed as Traditional Knowledge Digital Library (TKDL) as a repository of 2,50,000 formulations of various systems of Indian medicine.

Patenting

- A patent is an exclusive right granted for an innovation that generally provides a new way of doing something or offers a new technical solution to a problem.
- Conditions for Obtaining a Patent Protection
  → Novelty: Not part of 'State of the Art'.
  → Inventive step: Not obvious to a person skilled in the art.
  → Capable of industrial application: For the benefit of society.
- To Patent or Not to Patent an Invention
  → The inventor decides whether to patent or not.
- Inventions Eligible for Patenting
  → Patents may be granted for inventions/technologies in any field, ranging from a paper clip or ballpoint pen to a nanotechnology chip or a Harvard mouse (mouse with cancer genes).
- Non-Patentable Matters
  → Invention contrary to public morality
  → Mere discovery of a new form of a known substance
  → Arrangement or rearrangement
  → Literary, dramatic, musical, artistic work
  → Traditional knowledge (TK)

The World Intellectual Property Organization

- The World Intellectual Property Organization (WIPO), a specialized agency of the United Nations established in 1967, serves as the central hub for international cooperation in IP.
- WIPO's mission includes harmonizing patent laws and procedures across countries, which helps create a stable global IP environment, reducing complexities for international IP protection.
- Key instruments include the Patent Cooperation Treaty (PCT), the Madrid System for international trademark registration, and the Hague System for international industrial design registration.
- WIPO allows applicants to file a single international application that can be processed in multiple jurisdictions, streamlining the process and reducing costs.
- India engages in joint activities with foreign IP offices such as the USPTO, UKIPO, JPO, and EUIPO through Memorandums of Understanding (MoUs) and technical training programs.
- Beyond WIPO, the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) administered by the World Trade Organization (WTO) plays a central role in linking IP with international trade.